Data Inspection and Cleaning

Building Trust Before Analysis

A beginner-friendly guide to understanding, inspecting, and cleaning data before analysis.

Bad Data Creates Bad Decisions

Wrong Averages

Dirty values skew numeric summaries

Misleading Charts

Errors produce false visual patterns

Bad Models

Machine learning learns the wrong behavior

> Before trusting analysis, trust the data.

What Is Data Cleaning?

Cleaning does not mean making data perfect — it means making it reliable enough for the current purpose.

Fill Missing Values	Remove Duplicates
Standardize Text	Fix Data Types
Correct Invalid Values	Save Cleaned Version

Raw Data vs. Cleaned Data

Raw Data

- Original source version
- Useful for audit and review
- Should never be overwritten
- Allows cleaning to be repeated

Cleaned Data

- Working version after cleaning rules
- Used for statistics, charts, reports
- Saved separately from raw

Raw data is the source of truth. Cleaned data is the working version.

Dataset Anatomy

Rows, Columns, and Cells

Term	Simple Meaning	Example
Row	One record	One customer
Column	One variable	monthly_spend
Cell	One value	1200
Data Type	Kind of value	number, text, date

> Rows are records. Columns are variables. Cells are values.

What Is Data Inspection?

Inspection means checking what the dataset contains before making any changes.

> Inspection is how we learn the shape and health of the data.

Structure

Row count, column count, column names

Content

First rows, last rows, data types

Quality

Missing values, duplicates, unusual values

Consistency

Category spelling and formatting

Pandas Inspection Commands

The First Toolkit

Command	Purpose
`.shape`	Count rows and columns
`.head()`	View first rows
`.tail()`	View last rows
`.info()`	Check data types and non-null counts
`.describe()`	Summarize numeric columns
`.isna().sum()`	Count missing values
`.duplicated().sum()`	Count duplicate rows

These commands are not just code — they are questions we ask the dataset.

Shape

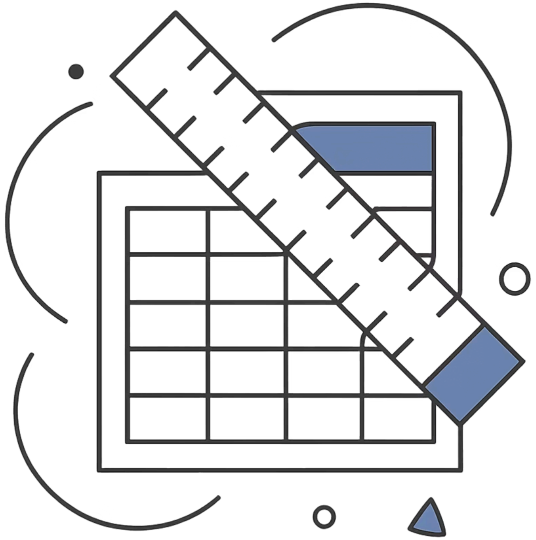
How Big Is the Dataset?

The shape returns `(rows, columns)`.

> Example: `(10, 7)` means 10 records and 7 variables.

- Confirms expected dataset size
- Detects accidental row loss
- Enables before/after comparison

> Always know how many rows you started with.



First Rows

Look at the Data Like a Human

See Example Values

Understand what each column actually contains

Spot Suspicious Values

Catch obvious problems before writing cleaning code

Verify Column Meaning

Confirm columns match expectations from the data dictionary

Manual inspection is not unprofessional. It is how we build intuition before automation.

Data Types

Does Each Column Have the Right Kind of Value?

Type	Simple Meaning	Example
integer	Whole number	5
float	Decimal number	1200.50
object/string	Text	Mumbai
datetime	Date or time	2026-06-15
boolean	True or False	True

> The same value can behave differently if the data type is wrong.

Missing Values

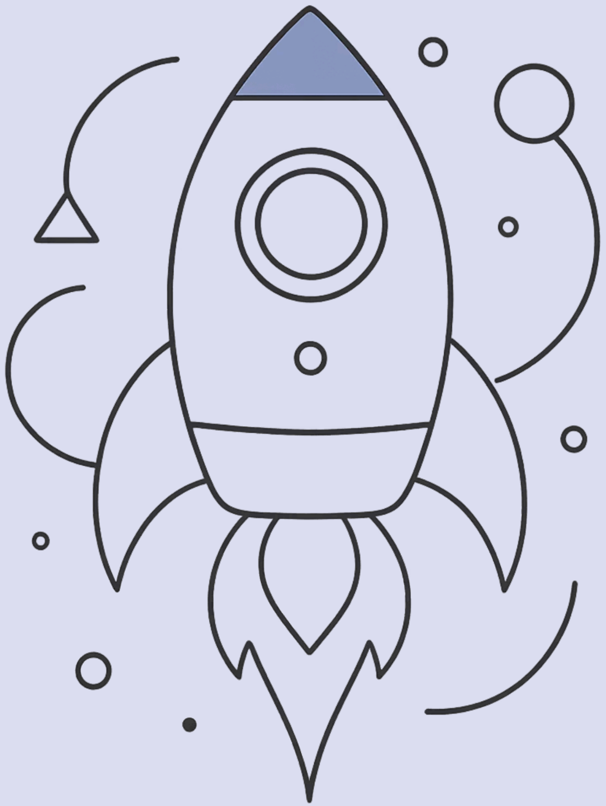
How They Appear

- Blank cells, `NaN`, `None`
- `null`, "Unknown", "Not available"

Why They Happen

- Data was not entered
- Value does not apply
- Source system failed
- File is incomplete

> Missing does not always mean mistake. It means investigate.



Missing Value Decisions

How Can We Handle Missing Data?

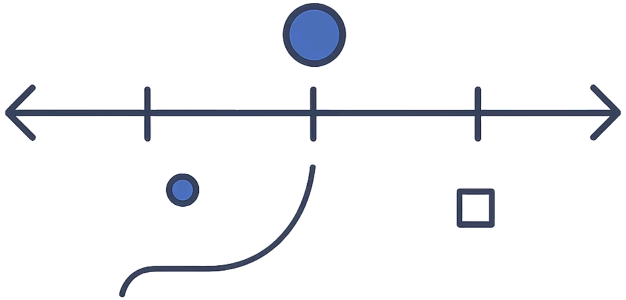
Action	Use When
Keep missing	Missingness itself may be meaningful
Fill numeric value	A suitable statistic makes sense
Fill category label	A label like "Unknown" is useful
Remove row	Very few rows affected and row is unusable
Remove column	Column is mostly missing and not important
Investigate source	Missingness is unexpected or risky

There is no universal best fix. The right choice depends on the column and the project question.

Numeric Missing Values

Filling Numbers Carefully

- Median — safer when values have outliers
- Mean — can be pulled by extreme values
- Zero — only when zero has real meaning
- Domain value — when business context applies



> ⓘ Example: `monthly_spend` missing → fill with median spend

> Never fill a number unless you can explain why that value makes sense.

Categorical Missing Values

Filling Text Categories

"Unknown"

Honest when we do not know the actual category

Most Common

Use the mode when a reasonable default exists

Domain Label

A business-specific label that fits the context

Keep Missing

When missingness itself carries meaning

Example: `city` missing → fill with "Unknown" — keeps the row and makes missingness visible.

Duplicate Rows

Repeated Records Can Distort Analysis

Overcounting

Same customer counted multiple times

Inflated Totals

Sums and averages become unreliable

Misleading Charts

Repeated evidence skews visualizations

> One real event should not accidentally count twice.

Duplicate Decisions

Remove or Keep?

Not every similar row is a duplicate. Ask:

- Is the entire row repeated?
- Could the same customer have multiple valid transactions?
- What does one row represent — customer, visit, or order?
- What does the data dictionary say?

Duplicate decisions depend on what one row represents.

Text Inconsistency

Same Meaning, Different Text

To a Human — All the Same

Mumbai	mumbai
MUMBAI	Mumbai

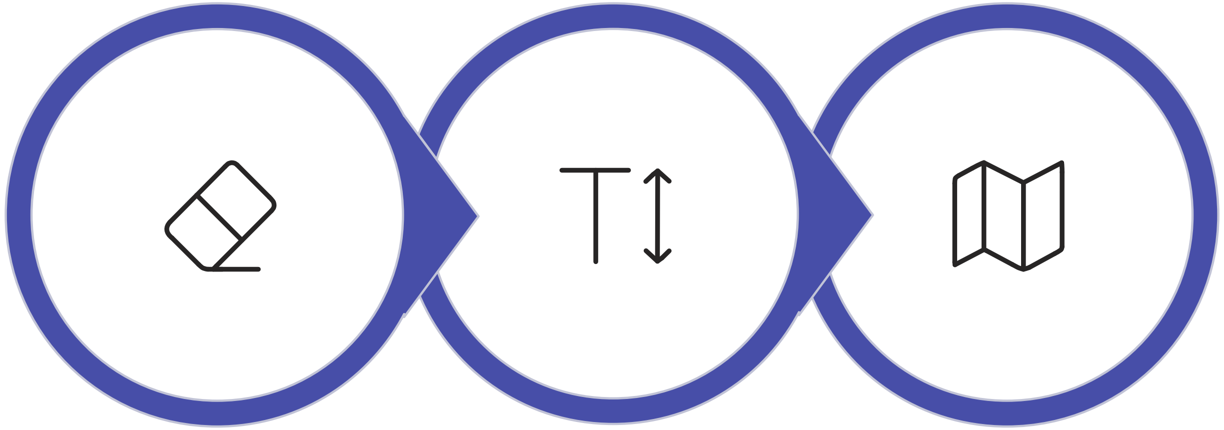
To Software — Four Different Categories

- Extra categories in group counts
- Wrong summaries per city
- Messy chart labels

> Software treats different spelling as different data.

Text Cleaning

Standardize Text Categories



Strip Spaces

Consistent Case

Map Variants

Use `.str.strip().str.title()` in Pandas to remove spaces and apply consistent capitalization. Text cleaning makes category counts meaningful.

Invalid Values

Values That Should Not Exist

Examples

- Monthly spend = `-500`
- Visits per month = `-3`
- Age = `300`
- Attendance = `150%`
- Signup month = `"Banana"`

Ask These Questions

- Is the value impossible?
- Is it a data entry issue?
- Can it be corrected?
- Should it be set to missing?
- Should the source be checked?

> Invalid values are not outliers. They are values that violate the rules of the data.

Outliers

Unusual Does Not Always Mean Wrong

Real Rare Cases

Important customers or one-time events

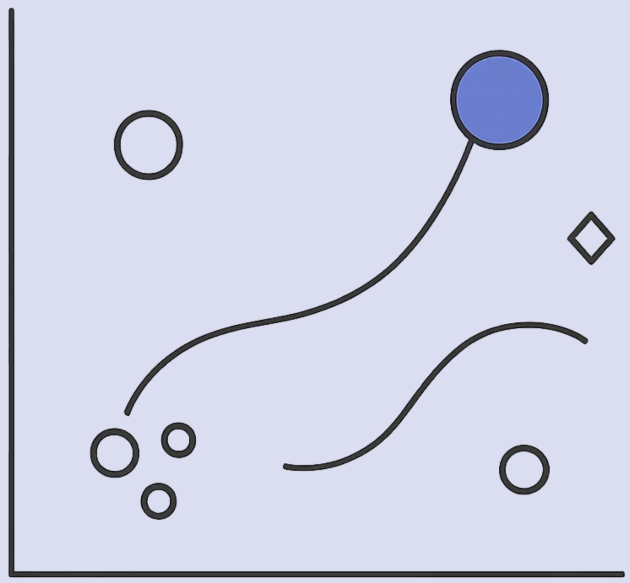
Data Entry Errors

Typos or system mistakes

System Errors

Upstream pipeline or export issues

⚠ Do not delete outliers automatically. First ask whether they are real, wrong, or important.



Invalid Value vs. Outlier

Different Problems Need Different Decisions

Situation	Example	Meaning	Action
Invalid value	spend = -500	Violates business rule	Fix, remove, or mark missing
Outlier	spend = 8000	Unusual but possible	Investigate before changing
Missing value	city is blank	Unknown value	Fill, keep, or investigate
Duplicate	Same row repeated	Repeated record	Remove if truly duplicate

> Wrong, missing, repeated, and unusual are different data problems.

Cleaning Rules

Document the Decision

What issue did we find?

Identify the specific problem and affected column

What rule did we apply?

State the cleaning action clearly

Why is the rule reasonable?

Justify the decision with logic or domain knowledge

Did we verify the result?

Confirm the issue is resolved after cleaning

Example: Issue: Missing `monthly_spend` → Rule: Fill with median → Verification: Missing count is now 0

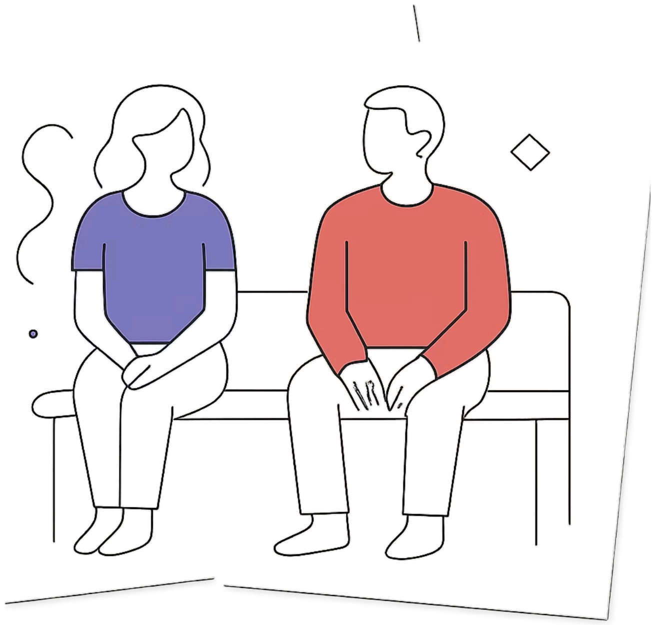
Copy Before Cleaning

Keep a Safe Working Version

```
clean_customers = raw_customers.copy()
```

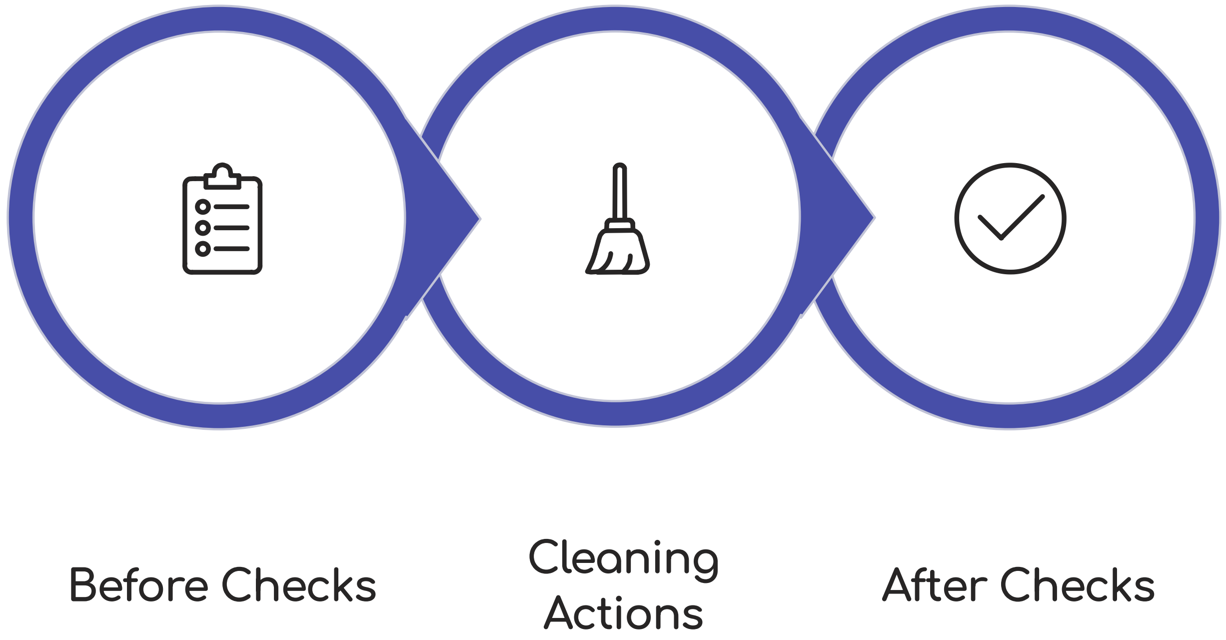
- Raw data stays unchanged
- Mistakes can be reviewed
- Cleaning can be repeated
- Before and after can be compared

> Do not destroy your original evidence.



Verify After Cleaning

Did the Cleaning Work?



After cleaning, rerun: row count, missing value count, duplicate count, data types, category values, numeric min/max, and sample rows.

Cleaning is incomplete until verification passes.

Row Count Awareness

Know What Changed

Cleaning Actions That Reduce Rows

Cleaning Is Iterative

One Pass Is Not Always Enough

A chart may reveal an impossible value.

A group summary may reveal inconsistent spelling.

Finding another issue later is normal — good workflows make it easy to go back.

Ethics and Privacy

Clean Data Responsibly

Protect Private Information

Mask or remove sensitive columns when appropriate

Ensure Fairness

Avoid removing rows from one group disproportionately

Document Assumptions

Cleaning decisions must be transparent and explainable

Data quality work is also responsibility work.

Common Beginner Mistakes

Cleaning Before Inspecting

Always inspect first

Overwriting Raw Data

Always copy before cleaning

Blind Missing Value Fills

Justify every fill decision

Auto-Deleting Outliers

Investigate before removing

Skipping Verification

Always check after cleaning

No Explanation

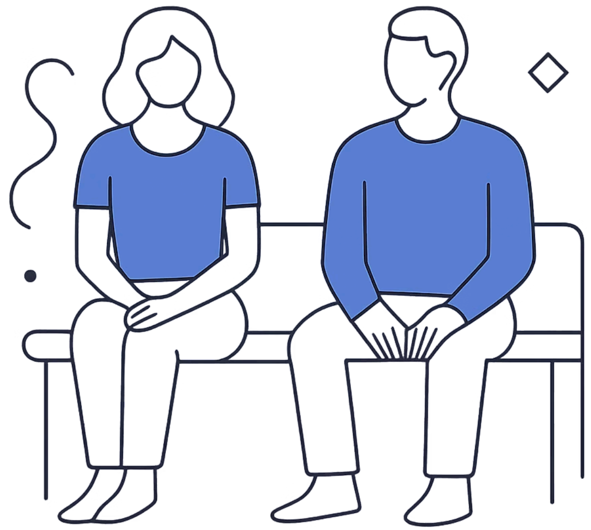
Document every cleaning choice

The goal is not to run many commands. The goal is to make trustworthy decisions about the data.

Before and After Thinking

Cleaning Should Have Evidence

Before	Action	After
Missing `monthly_spend` = 1	Fill with median spend	Missing `monthly_spend` = 0



> If you cannot show what changed, the cleaning is hard to trust.

Discussion Prompt

Think Like an Analyst

> If `monthly_spend` is missing for one customer, what should we do?

- Fill with Mean
- Fill with Median
- Leave Missing
- Remove the Row
- Ask the Source

All of these can be reasonable in different situations. The important skill is explaining the decision.

Mini Case

Cleaning Decisions in Context

Customer data has: one missing city, one missing monthly spend, one duplicate row, and one negative monthly spend.

Missing City

Fill with "Unknown"

Missing Monthly Spend

Fill with median spend

Duplicate Row

Remove exact duplicate

Negative Spend

Replace with reasonable value or investigate source

The cleaning plan should be simple, explainable, and checked.

Section Summary

What We Learned

Inspect First

Shape, head, info, missing counts, and duplicate checks are the first tools

Decide Deliberately

Missing values, duplicates, text, and outliers each require a reasoned decision

Document and Verify

Cleaning rules should be documented and results confirmed

Save Separately

Cleaned data is saved apart from raw data

> Trustworthy analysis starts with trustworthy data.

Key Principles to Carry Forward

Inspect Before You Clean

Understanding the data comes before fixing it

Clean with Decisions, Not Guesses

Every cleaning action needs a reason

Verify Every Change

Cleaning is incomplete until the result is confirmed

Preserve the Original

Raw data is the source of truth — never overwrite it